Episode 7

Documentation & Architecture Artifacts

"If it's not documented, it doesn't exist"

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Think Like an Architect

المعمارية بـبساطة



WHERE ARE WE?

Ep 1–2: Foundations	Ep 3–4: Design	Ep 5–6: Patterns	Ep 7: Docs ★

Think Like an Architect

1. Introduction to Solution Architecture ✅
2. Think Like an Architect ✅
3. Requirements Engineering ✅
4. Design Principles & Architecture Patterns ✅
5. Data & Integration Architecture ✅
6. Security by Design ✅



Think Like an Architect — Episode 7

Documentation& Architecture Artifacts

YOU DESIGNED IT. CAN OTHERS UNDERSTAND IT?



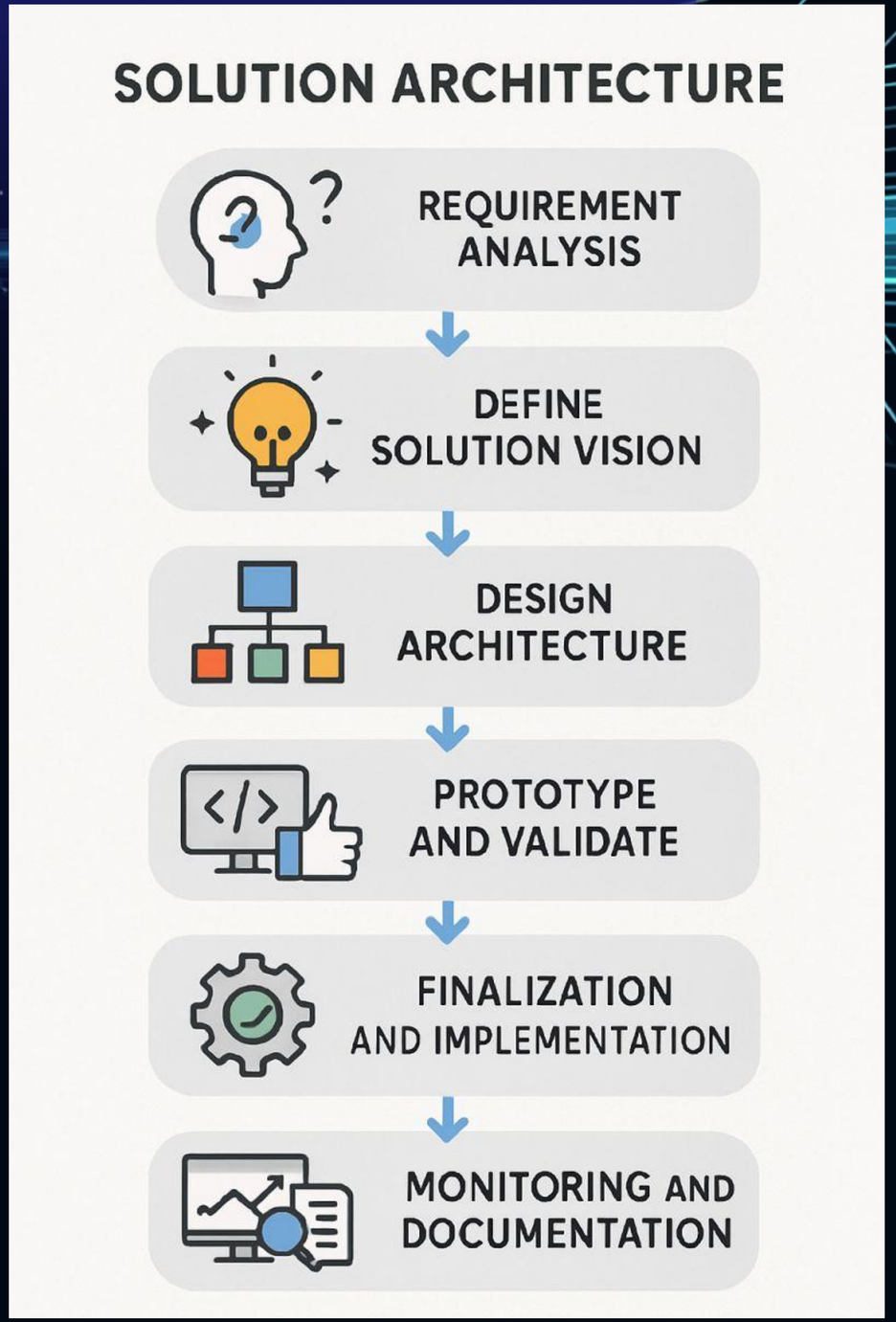
SOLUTION IMPLEMENTATION NOT **THE END**

Why is documentation important?



Communicate	Align	Decide	Preserve
			

FROM REQUIREMENTS TO ARCHITECTURE



DOCUMENTATION LANDSCAPE

Category	Document Type	Key Components
Business	BRD	Scope, Goals
Solution	HLD	Blueprint, Overview
Technical	LLD	Network, Security, Data
Operations	Runbooks	Deployment, Monitoring

BRD → HLD → LLD

Business View

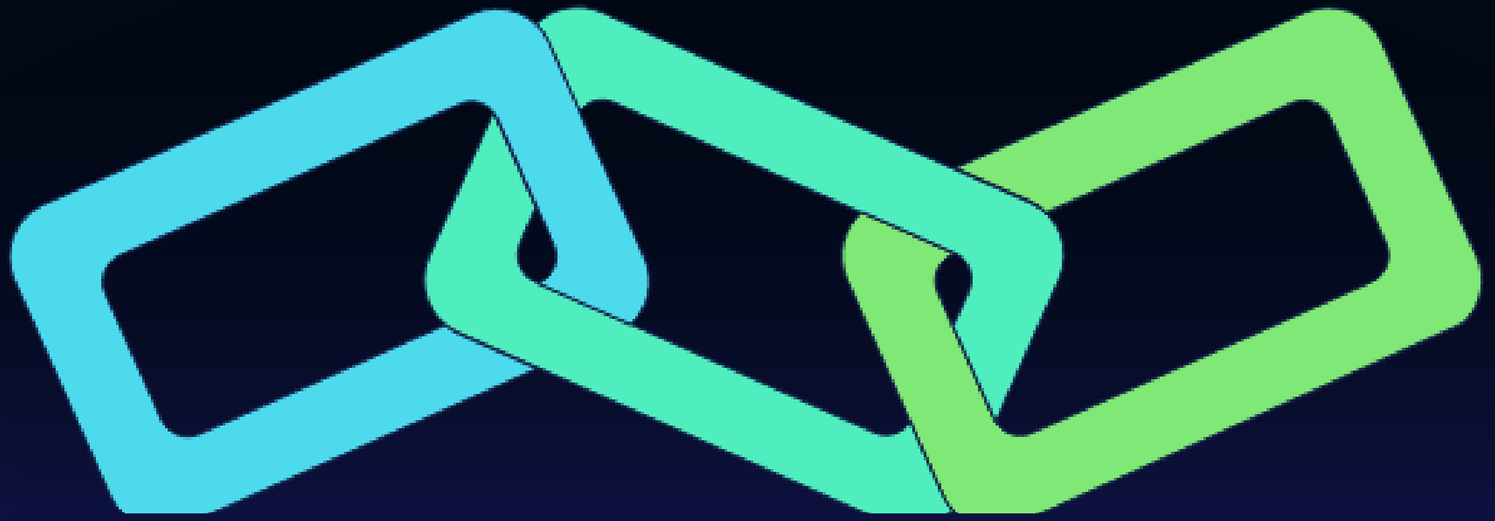
Focuses on the business's needs and goals

Solution View

Outlines how the solution will function

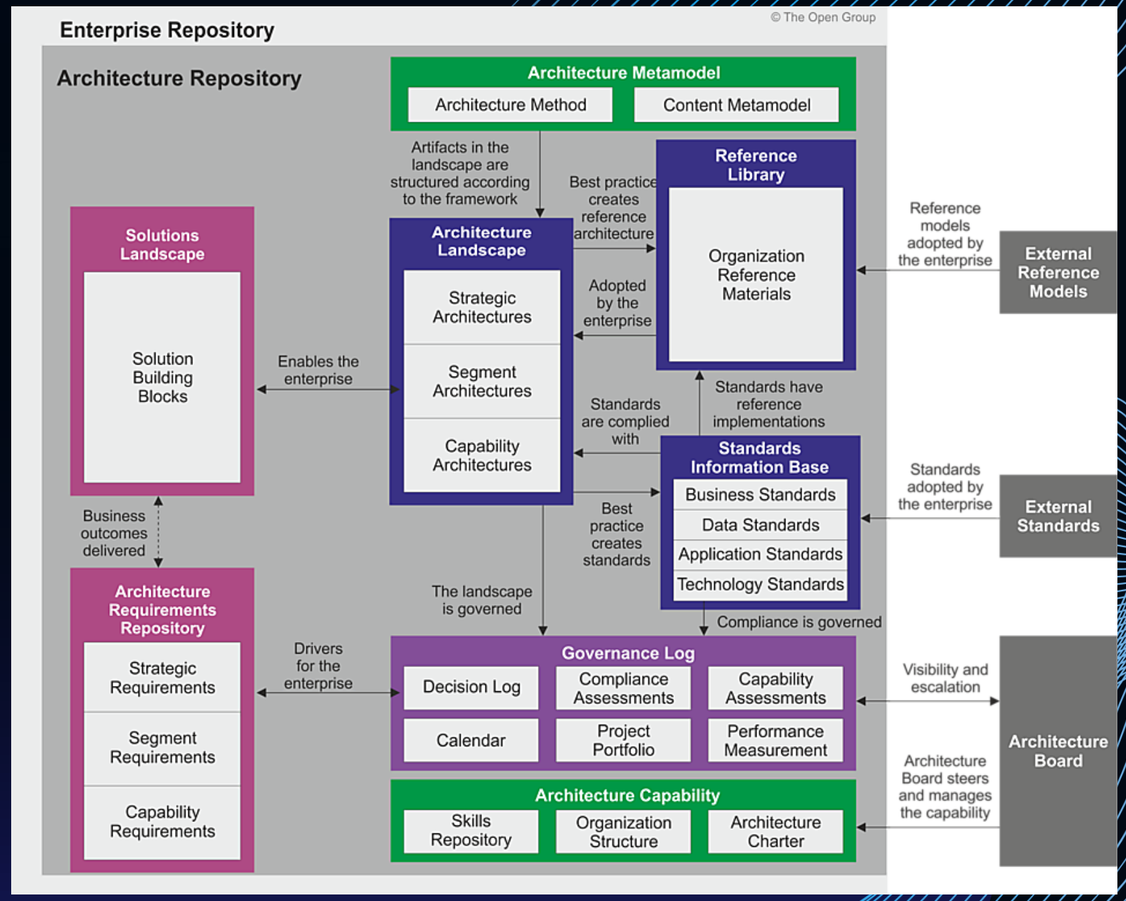
Technical View

Details the implementation process and technologies



WHAT IS AN **ARCHITECTURE ARTIFACT** AND **ARCHITECTURE REPOSITORY**?





Architecture Artifact Tools

GitHub / Azure DevOps

Version Control & Docs-as-Code

Store ADRs, diagrams & docs alongside code

SharePoint

Enterprise Knowledge Base

Centralized artifact storage & access control

Confluence

Wiki & Living Docs

Team wikis, HLD/LLD pages & runbooks

Notion

Modern Documentation

Flexible docs & architecture knowledge mgmt

Azure DevOps Wiki

Developer Portal

Integrated wiki tied to your project & pipelines

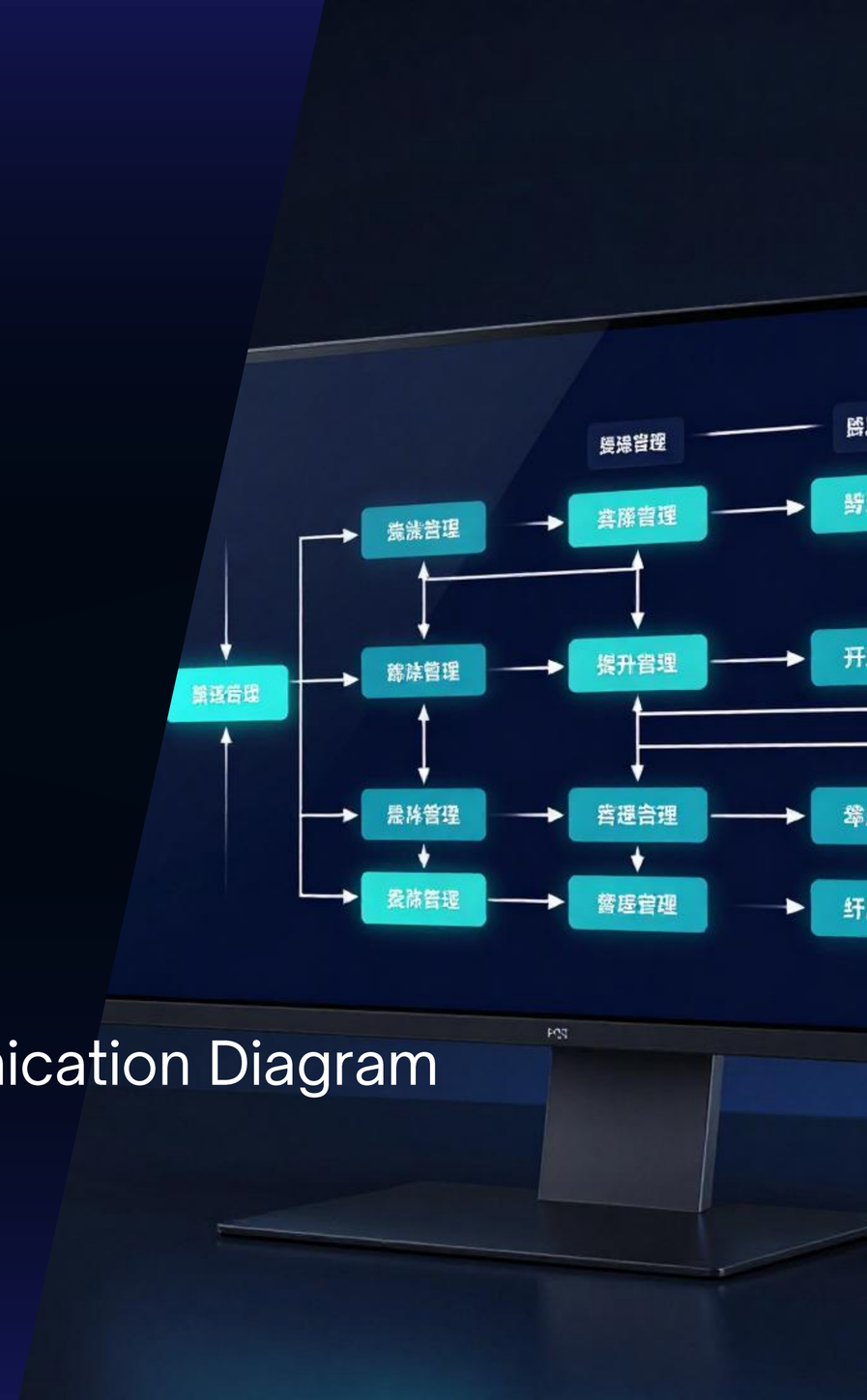
Backstage (Spotify)

Internal Dev Portal

Centralized catalog for services, docs & specs

ARCHITECTURE DIAGRAMS

- High-level diagram (HLD)
- Low-level diagram (LLD)
  - Network Architecture Diagram
  - Data Flow Diagram
  - Sequence Diagram
  - Integration / Application Communication Diagram
  - Security Architecture Diagram
  - Disaster Recovery Diagram



CHOOSING THE RIGHT DIAGRAM



business user



Security



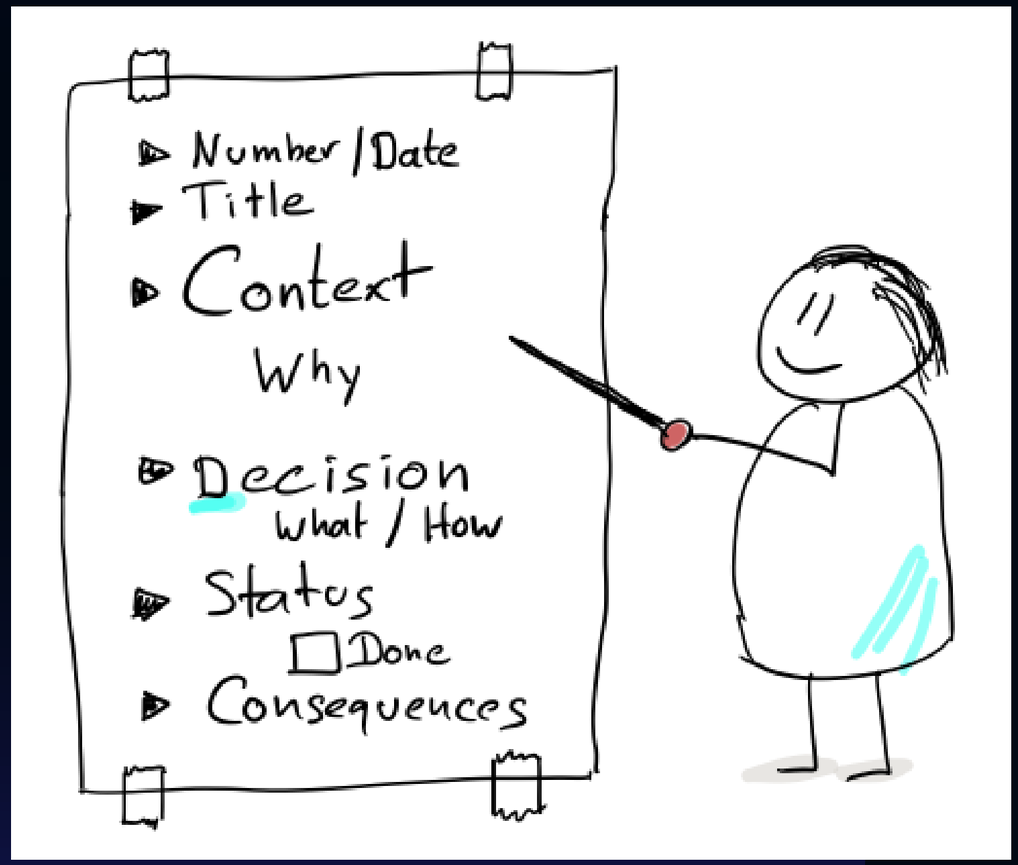
Data Enginer



Technical

ARCHITECTURE DECISION RECORD (ADR)

REAL EXAMPLE — ADR-007: API Gateway Selection
Context: E-Commerce platform requires a unified entry point for 12 microservices. Current direct service calls cause auth duplication and poor observability.
Decision: Adopt AWS API Gateway with Lambda authorizers as the single ingress layer.



DECISION LOG VS ADR



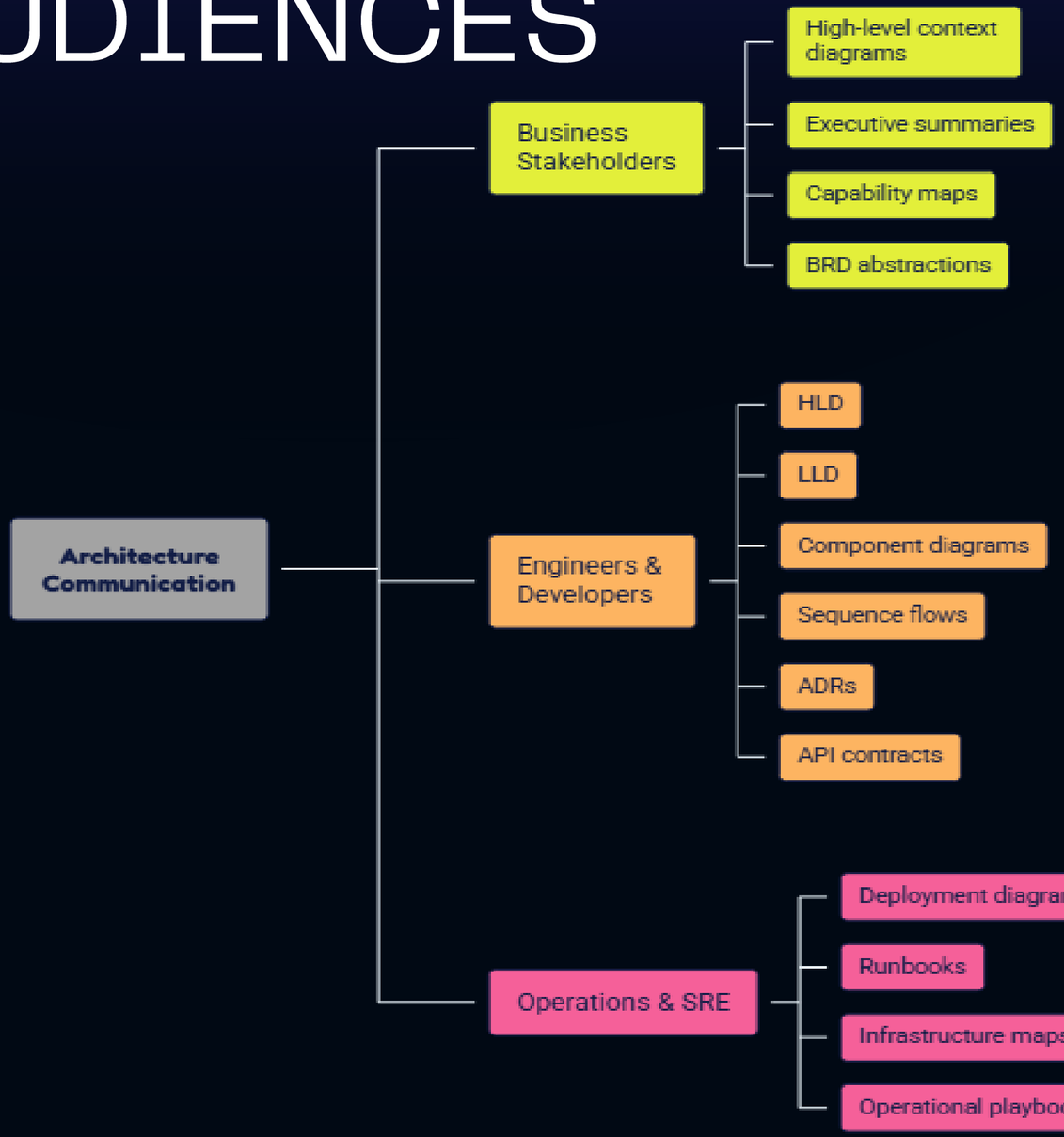
Decision Log

Focuses on the "What" of the decision, detailing the specific choice made.

ADR

Addresses the "Why" of the decision, explaining the rationale behind the choice.

ONE ARCHITECTURE, DIFFERENT AUDIENCES



COMMON DOCUMENTATION MISTAKES



❌ Outdated diagrams that no longer reflect reality
❌ Excessive detail that obscures the big picture (HLD)
❌ Missing decisions — the "**why**" is never recorded

❌ Unclear ownership — no one maintains the docs
❌ Documenting without a defined audience or purpose
❌ Treating documentation as a one-time deliverable

THANK YOU

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